MEHEDI HASAN

Python, Automation & Full-Stack Tooling Developer | Undergraduate CSE Researcher

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EXECUTIVE SUMMARY

Systems-oriented Computer Science & Engineering undergraduate and Undergraduate Research Assistant at the **FEC CSE Research Lab** with deep hands-on experience building full-stack web applications, OS-level Windows desktop overlays, and Manifest V3 browser automation extensions. Proven track record of designing high-concurrency real-time streams, hybrid AI/EXIF verification pipelines, and sub-millisecond SQL aggregation architectures. Available for Software Engineering Internships, Research Collaborations, Freelance Projects, and Full-Time Roles.

EDUCATION & ACADEMIC RESEARCH

Faridpur Engineering College (FEC) — B.Sc. in Computer Science & Engineering (CSE) *Faridpur, Bangladesh | Active / Present*

- **Core Focus:** Algorithms, Data Structures, Systems Engineering, Database Architecture, and Full-Stack Software Tooling.

FEC CSE Research Lab — Undergraduate Research Assistant *Faridpur Engineering College | Present*

- Conducting applied Computer Science & Engineering research in collaboration with faculty advisors and lab peers; actively preparing research manuscripts and experimental pipelines for upcoming peer-reviewed publication.

TECHNICAL SKILLS & TOOLING

Languages:	Python (Primary), C, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
Frameworks & Web:	Django, Flask, SQLAlchemy ORM, Node.js, Tailwind CSS, Leaflet.js (GIS), Server-Sent Events (SSE)
Systems & Desktop:	Windows Win32 API (ctypes, User32/Shcore), Tkinter GUI, Chrome/Edge Extension APIs (Manifest V3), Pillow (PIL), ReportLab
Databases & DevOps:	PostgreSQL, SQLite (Composite Indexing & Bulk Aggregations), Git, GitHub Actions CI/CD, Vercel, Linux

FEATURED ENGINEERING & FULL-STACK PROJECTS

Fix_my_city — Bangladesh Civic Infrastructure Issue Tracker & GIS Platform *Live Demo | GitHub Repo*

Stack: Python, Django, Leaflet.js (GIS), ReportLab (PDF), EXIF/Binary Image Analysis, HTML5/CSS3/JS

- Engineered a two-tier civic-tech platform (Citizen Reporter & Municipal Authority Triage Console) with immutable audit trails (**StatusHistory**) across Dhaka municipal wards (DNCC & DSCC).
- Built a hybrid AI-generated image & spam verification pipeline combining Camera Hardware EXIF fast-path validation with a binary header/noise detector blocking synthetic uploads (Midjourney/DALL-E).
- Automated statutory PDF petition generation under the Local Government Act 2009 & RTI Act 2009 using **ReportLab** with GPS watermarks and interactive multi-layer **Leaflet.js** Nominatim reverse geocoding.

Meal_management_web_app (Plate & Spoon) — Real-Time Hostel Meal & Financial Auditing System *Live App | GitHub Repo*

Stack: Python, Flask, SQLAlchemy ORM, PostgreSQL/SQLite, Server-Sent Events (SSE), Vanilla JS, Tailwind CSS

- Architected a thread-safe real-time **Server-Sent Events (SSE)** streaming engine (**MessageAnnouncer** with **threading.Lock()**) for non-blocking live group chat and instant financial notifications.
- Optimized database performance via composite indexes and bulk SQL aggregations (**get_bulk_user_monthly_stats**), cutting dashboard load complexity from 100+ N+1 queries down to 3–4 sub-millisecond queries.
- Designed a dynamic financial auditing engine calculating live per-meal rates, member deposits, shared expenses, role-based slot locks, and net balances with seamless manager role transfers.

Convert_php_to_usdt (calc-php-converter) — Desktop Currency & OCR Calculator Overlay *Release v4.0.0 | GitHub Repo*

Stack: Python, Tkinter GUI, Pillow (PIL), Windows Win32 API (ctypes: User32 & Shcore)

- Built a frameless, always-on-top Windows desktop overlay utility for instant PHP ↔ USDT currency conversion, live math evaluation, dynamic UI scaling, and integrated quick notes.

- Integrated OS-level Win32 APIs via **ctypes** for global system hotkeys (**Ctrl+/** to summon, **Ctrl++/-** for live transparency) and High-DPI monitor scaling alongside a Google Lens OCR screen-region snipping tool.

Safe-Seller-Keep-Alive — Stealth Marketplace Session Manager (Manifest V3) *Release v1.3.0* | *GitHub Repo*

Stack: JavaScript (ES6+), Chrome & Edge Extension Manifest V3 (chrome.alarms, chrome.power, Service Workers)

- Developed a background Manifest V3 extension that keeps G2G & U7BUY seller accounts active in a single background tab using randomized jitter intervals (600s–1000s) without stealing window focus.
- Engineered hardware wake-lock protection (**chrome.power** preventing Windows S0 Modern Standby), automatic timer recovery after PC restarts, and Cloudflare challenge / logout detection safeguards.

AVAILABILITY & COLLABORATION

Open to **Software Engineering Internships**, **Undergraduate Research Collaborations**, **Freelance Tooling & Web Projects**, and **Full-Time Engineering Opportunities**. Direct contact via Email (mhbhuiyan2004@gmail.com), LinkedIn (mehedi-hasan-a8bb362a1), Telegram (@Mh_13585), or Discord (**mh_13585**).